

Unit-6: Working with Files

QB Solution

Q:503 Write a function cust_data() to ask user to enter their names and age to store data in customer.txt file.

```
In [30]: def cust_data():  
         name=input("Enter Name: ")  
         age=input("Enter Age: ")  
         f=open("Customer.txt","a")  
         f.write(name+"-"+age+"\n")  
         f.close()  
  
cust_data()  
cust_data()
```

```
Enter Name: Python  
Enter Age: 10  
Enter Name: Java  
Enter Age: 20
```

Q:504 Write a python program to create and read the city.txt file in one go and print the contents on the output screen.

```
In [32]: f=open("City.txt","w+")  
         f.write("Python Programming\n")  
         f.write("Python Programming is easy\n")  
         f.write("Python Programming is fun\n")  
         f.seek(0)  
         print(f.read())  
         f.close()
```

```
Python Programming  
Python Programming is easy  
Python Programming is fun
```

Q:505 Write a function count_lines() to count and display the total number of lines from the file. Consider the following lines for the file – friends.txt.

- Friends are crazy, Friends are naughty !
- Friends are honest, Friends are best !
- Friends are like keygen, friends are like license key !
- We are nothing without friends, Life is not possible without friends !

```
In [33]: def count_lines():  
         f=open("Friend.txt","r")  
         line=len(f.readlines())  
         print("No of line:",line)  
         f.close()  
count_lines()
```

No of line: 4

Q:506 Write a function display_oddLines() to display odd number lines from the text file. Consider the following lines for the file – friends.txt.

- Friends are crazy, Friends are naughty !
- Friends are honest, Friends are best !
- Friends are like keygen, friends are like license key !
- We are nothing without friends, Life is not possible without friends !

```
In [34]: def display_oddLines():  
         f=open("Friend.txt","r")  
         line=f.readlines()  
         for i in range(0,len(line),2):  
             print(line[i])  
         f.close()  
display_oddLines()
```

Friends are crazy, Friends are naughty !

Friends are like keygen, friends are like license key !

Q:507 Write a Python program to read a text file and do following: 1. Print no. of words 2. Print no. statements

```
In [38]: f=open("Test.txt","r")  
         line=f.readlines()  
         w=0  
         for word in line:  
             w+=len(word.split())  
  
         print("Line:",len(line))  
         print("Words:",w)  
         f.close()
```

Line: 3

Words: 10

Q:508 Write a python program that reads a text file and changes the file by capitalizing each character of file.

```
In [39]: f=open("Test.txt","r+")
data=f.read()
new=data.upper()
f.seek(0)
f.write(new)
f.close()
```

Q:509 Write a Python program to copy the contents of a file to another file.

```
In [40]: f1=open("Test.txt","r")
f2=open("Test2.txt","w")
data=f1.read()
f2.write(data)
f1.close()
f2.close()
```

Q:510 Write a python program to read line by line from a given files file1 & file2 and write into file3.

```
In [8]: f1=open("Friend.txt")
        f2=open("Test.txt")
        f3=open("File3.txt","w")
        data1=f1.readlines()
        data2=f2.readlines()
        flage=True

        if len(data1)>len(data2):
            l=len(data2)
        else:
            l=len(data1)
            flage=False

        for i in range(l):
            f3.write(data1[i])
            f3.write(data2[i])

        if flage:
            x=data1[l:]
            f3.writelines(x)
        else:
            x=data2[l:]
            f3.writelines(x)

        f1.close()
        f2.close()
        f3.close()
```

Q:511 Write python program to count the number of lines in a file.

```
In [9]: f=open("Test.txt","r")
        line=f.readlines()
        print("Line:",len(line))
        f.close()
```

Line: 3

Q:512 Write a python program to search for a string in text files.

```
In [13]: u=input("Enter String: ")
f=open("File3.txt","r")
data=f.read()
i=data.find(u)
if i== -1:
    print("String not Found.")
else:
    print("String found at position:",i)
f.close()
```

Enter String: crazy
String found at position: 12

Q:513 Write a “pager” program. Your solution should prompt for a filename, and display the text file 25 lines at a time, pausing each time to ask the user to enter the word “continue”, in order to show the next 25 lines or enter the word “stop” to close the file.

```
In [3]: file=input("Enter File Name:")
f=open(file)
data=f.readlines()
start=0
size=25
total_lines=len(data)
while start < total_lines:
    for line_num in range(start, min(start + size, total_lines)):
        print(data[line_num].rstrip())

    u=input("Continue or Stop?")

    if u.lower()=="stop":
        break
    start+=size

print("End of File")
f.close()
```

```
Enter File Name:Friend.txt
Friends are crazy, Friends are naughty !
Friends are honest, Friends are best !
Friends are like keygen, friends are like license key !
new We are nothing without friends, Life is not possible without friends !
! SDNEIRF TUOHTIW ELBISSOP TON SI EFIL ,SDNEIRF TUOHTIW GNIHTON ERA EW WEN
! YEK ESNECIL EKIL ERA SDNEIRF ,NEGYEK EKIL ERA SDNEIRF
! TSEB ERA SDNEIRF ,TSENOH ERA SDNEIRF
! YTHGUAN ERA SDNEIRF ,YZARC ERA SDNEIRF
! SDNEIRF TUOHTIW ELBISSOP TON SI EFIL ,SDNEIRF TUOHTIW GNIHTON ERA EW WEN
Friends are crazy, Friends are naughty !
Friends are honest, Friends are best !
Friends are like keygen, friends are like license key !
Friends are crazy, Friends are naughty !
Friends are honest, Friends are best !
Friends are like keygen, friends are like license key !
! TSEB ERA SDNEIRF ,TSENOH ERA SDNEIRF
! YTHGUAN ERA SDNEIRF ,YZARC ERA SDNEIRF
Friends are crazy, Friends are naughty !
Friends are honest, Friends are best !
Friends are like keygen, friends are like license key !
new We are nothing without friends, Life is not possible without friends !
! SDNEIRF TUOHTIW ELBISSOP TON SI EFIL ,SDNEIRF TUOHTIW GNIHTON ERA EW WEN
! YEK ESNECIL EKIL ERA SDNEIRF ,NEGYEK EKIL ERA SDNEIRF
! TSEB ERA SDNEIRF ,TSENOH ERA SDNEIRF
! YTHGUAN ERA SDNEIRF ,YZARC ERA SDNEIRF
Continue or Stop?Stop
End of File
```

Q:514 Using a file which consists of multiple statements, find all the words from the file that can be made from all the characters of given user's string.

Note: If user enters same characters multiple times in a string, then word from file will only be eligible for output if it contains that character for same or more number of times in it. (if file have apple, greenapple and user's string is 'aepe' then output will only be greenapple. apple is not eligible as it contains e for 1 time only.

```
In [15]: user=input("Enter Characters: ")
with open('Test2.txt', 'r') as file:
    data=file.read().split()
    for word in data:
        for char in user:
            if char not in word:
                break
            if user.count(char) > word.count(char):
                break
        else:
            print(word)
```

Enter Characters: eee
greenapple

Q:515 Write a Python program to reverse the content of a one file and store it in second file and also convert content of second file into uppercase and store it in third file and also count number of Vowels in third file and also print only 2nd line from the content of third file.

Examples:

If data file one contains the following data:

- Friends are crazy, Friends are naughty !
- Friends are honest, Friends are best !

Output 1:

- ! tseb era sdneirF ,tsenoh era sdneirF
- ! ythguan era sdneirF ,yzarc era sdneirF

Output 2:

- ! TSEB ERA SDNEIRF ,TSENOH ERA SDNEIRF
- ! YTHGUAN ERA SDNEIRF ,YZARC ERA SDNEIRF

Output 3:

- Vowels = 22

Output 4:

- ! YTHGUAN ERA SDNEIRF ,YZARC ERA SDNEIRF

In [4]:

```
# Reverse the content of input file and store it in reversed file
with open('Friend.txt', 'r') as file:
    content = file.read()
    with open("reversed.txt", 'w') as rev_file:
        rev_file.write(content[::-1])

# Convert content of reversed file to uppercase and store it in uppercase file
with open("reversed.txt", 'r') as rev_file:
    content = rev_file.read().upper()
    with open("uppercase.txt", 'w') as up_file:
        up_file.write(content)

# Count number of vowels in the uppercase file
vowels = 'aeiouAEIOU'
c=0
with open("uppercase.txt", 'r') as uppercase_file:
    content = uppercase_file.read()
    for char in content:
        if char in vowels:
            c=c+1
    print(c)

# Print only the second line from the content of the uppercase file
with open("uppercase.txt", 'r') as uppercase_file:
    lines = uppercase_file.readlines()
    if len(lines) >= 2:
        print("Second line from the file:")
        print(lines[1])
```

63

Second line from the file:

! SDNEIRF TUOHTIW ELBISSOP TON SI EFIL ,SDNEIRF TUOHTIW GNIHTON ERA EW WEN

Q:516 Write a python program to extract a list of all four-letter words that start and end with the same letter from a given text file.

In [8]:

```
f=open("Test.txt","r")
data=f.read().split()
for word in data:
    if (len(word)==4 and word[0]==word[-1]):
        print(word)
f.close()
```

test
PyyP
area

Q:517 Write a python program to read a text file “Story.txt” and print only word starting with ‘I’ in reverse order.

Example: If value in text file is : ‘INDIA IS MY COUNTRY’

Output will be: ‘AIDNI SI MY COUNTRY’

```
In [26]: f=open("Test.txt","r")
data=f.read().split()
new=""
for word in data:
    if word[0]=="I":
        new+=word[::-1]+" "
    else:
        new+=word+" "
print(new)
f.close()
```

AIDNI SI MY COUNTRY

Q:518 Write a Python program to count words, characters and spaces from text file.

Input:

- Python is a Easy Subject
- OOPs is One of the most
- interesting Topic

Output:

- No of space: 10
- No of word: 13
- No of character: 64

```
In [22]: f=open("Test.txt","r")
file=f.read()
data=file.split()
w=len(data)
c=0
for i in data:
    c+=len(i)
s=file.count(" ")

print("No of space:",s)
print("No of word:",w)
print("No of character:",c+s)

f.close()
```

No of space: 10
No of word: 13
No of character: 64

Q:519 File Filtering. write all lines of a file1, except those that start with a pound sign (#), the comment character for Python to file2. And display data of file2.

Text file1 content:

- Friends are crazy, Friends are naughty !
- #Friends are honest, Friends are best !
- Friends are like keygen, #friends are like license key !
- We are nothing without friends, Life is not possible without friends !

Text file2 should be:

- Friends are crazy, Friends are naughty !
- Friends are like keygen,
- We are nothing without friends, Life is not possible without friends !

```
In [10]: with open("Friend.txt","r") as f1, open("FilterFile.txt","w") as f2:
        for i in f1:
            if i[0]=="#":
                continue
            f2.write(i)
```

Q:520 Write a python program to accept string/sentence from user till the user enters "END". Each string/sentence entered by user should be a newline in file. Save all the lines in file and display only those lines which begin with capital letter.

Example:

- Enter Something (for quit enter END):Hi Friends
- Enter Something (for quit enter END):how are you all
- Enter Something (for quit enter END):I am fine
- Enter Something (for quit enter END):hope you all are fine
- Enter Something (for quit enter END):END
- The Line started with Capital Letters:
- Hi Friends
- I am fine

```
In [16]: f=open("User.txt","w+")
while True:
    u=input("Enter Something (for quit enter END): ")
    if u=="END":
        break
    else:
        f.write(u+"\n")

f.seek(0)

for line in f:
    if line[0].isupper():
        print(line[:-1])

f.close()
```

```
Enter Something (for quit enter END): HI
Enter Something (for quit enter END): Hello
Enter Something (for quit enter END): java
Enter Something (for quit enter END): ai
Enter Something (for quit enter END): ml
Enter Something (for quit enter END): Python
Enter Something (for quit enter END): END
HI
Hello
Python
```

Q:521 Write a program to compare two text files. If they are different, give the line and column numbers in the files where the first difference occurs.

Example:

File 1: python1.txt

- Friends are crazy, Friends are naughty !
- Friends are honest, Friends are best !
- Friends are like keygen, friends are like license key !
- new We are nothing without friends, Life is not possible without friends !

File 2: python2.txt

- Friends are crazy, Friends are naughty !

- Friends 6re honest, Friends are best !
- Friends are like keygen, friends are like license key !
- new We are nothing without friends, Life is not possible without friends !

Output:

line number 2 colNo: 9

```
In [13]: f1=open("Friend.txt")
f2=open("Friend2.txt")

data1=f1.read()
data2=f2.read()

line,col=1,1

if data1==data2:
    print("Both files are Same")
else:
    l=min(len(data1),len(data2))
    for i in range(l):
        if data1[i]==data2[i]:
            col+=1
        else:
            print("Line No: ",line,"Col: ",col)
            break

    if data1[i]=="\n":
        line+=1
        col=1
f1.close()
f2.close()
```

Line No: 2 Col: 9

Q:522 Write a Python program to count the frequency of words in a file**Example:****File 1: python1.txt**

- Friends are crazy, Friends are naughty !
- Friends are honest, Friends are best !
- Friends are like keygen, friends are like license key !
- new We are nothing without friends, Life is not possible without friends !

Output:

Friends - 5,are - 7,crazy, - 1,naughty - 1,! - 4,honest, - 1,best - 1,like - 2,keygen, - 1,friends - 2,license - 1,key - 1,new - 1,We - 1,nothing - 1,without - 2,friends, - 1,Life - 1,is - 1,not - 1,possible - 1,

```
In [4]: f=open("Friend.txt")
data=f.read().split()
d={}
for i in data:
    if i in d:
        d[i]+=1
    else:
        d[i]=1

for i in d:
    print(i,"-",d[i],end=" ",)

f.close()
```

Friends - 5, are - 7, crazy, - 1, naughty - 1, ! - 4, honest, - 1, best - 1, like - 2, keygen, - 1, friends - 2, license - 1, key - 1, new - 1, We - 1, no thing - 1, without - 2, friends, - 1, Life - 1, is - 1, not - 1, possible - 1,

In []: